

Patrick Thesis Notes

Title

The current title is descriptive of the method, but does not illustrate the main findings. I think this is okay for a thesis, but just wanted to highlight this as for a research paper I'd recommend a title that reflects the key finding of the study. Thesis titles are very personal thing though, so I'd leave it up to you. However, double check that the title makes sense semantically and grammatically. Personally, I would not "use" a hypothesis. Perhaps "test" could be a better word? Or "applying" as you use in the abstract. For example:

"Testing the spectral variability hypotheses for biodiversity monitoring in the Arctic tundra using hyperspectral images".

"Applying the spectral variability hypotheses for biodiversity monitoring in the Arctic tundra using hyperspectral images".

You could also consider using biodiversity "assessment(s)" rather than "monitoring", which to me implies more of a time-series / change over time component. I think both work equally well. Again, these are just suggestions and I would leave the final choice of title up to you.

Imprint

Check my details - it currently states that I am at the University of Cambridge 😊

Abstract

I enjoyed reading the abstract! It was concise and to the point - great job 😊

General note on passive vs. active voice

You often chose to write in the passive voice "X was done" rather than "I did X" or "X did Y" (active voice). That is okay, but I just wanted to highlight that scientific writing is moving towards the active voice as it's easier to follow and understand. It also avoids creating a false sense of objectivity that has been prevailing in the natural sciences. It's okay to mix both as well, most papers nowadays lean towards using more active voice.

Acknowledgements

Thank you for the kind words! Always one of the nicest parts to read for us. So great to hear your family was so supportive and I am sure your kind words will mean a lot to them.

Abbreviations

Very comprehensive and clear. Nice overview!

Introduction

Again, this was very nicely written. Good funnel structure (from big picture down to the narrow thesis topic). Good use of literature and citations. Good length of section overall.

Two small things:

1. The paragraphs seemed a bit unbalanced in terms of length. Some were longer others were shorter. As this is a thesis not a paper for submission to a journal, I think you have a bit more flexibility, but if you have time consider adding a little bit more "meat" to paragraphs #1 (big picture) and #4 (underrepresentation of tundra in hyperspectral studies).
2. I'd suggest being perhaps a bit more careful with citing Marie-Lous thesis, especially when it comes to big picture statements such as "There is a need for scalable approaches in Arctic monitoring". If you already have one or two papers published in a journal, I would probably not cite her thesis as she actually did not do research into whether these approaches are needed or now. A thesis is a great citation if no other sources are available - there are definitely a few cases in your intro where this is true.

Method

Nice work overall, but perhaps some clarifications needed. See in text comments.

Generally, a bit more detail on parameters etc was provided than what we would do in a publication (just for your interest). There we usually report only the most critical parameters and then publish the code with all the details. However, I think in the case of a thesis the level you chose is okay. However, the way that the "code" is rendered in text is very large and it kinda disrupts the reading flow. I suggest either using the same font and size for code and text, or reducing the size of the monospaced code font.

Will you make your code repository available? If yes, you could add a section to the method providing the GitHub link and perhaps an archived release published on Zenodo (and minted with a DOI?).

Results

Looks good overall, but I think it would benefit from a bit of a tidy up. See in-text comments and suggestions. You could also tidy up the figures a bit more (remove redundant legends, make titles and labels less crammed etc.).

For the, spectral species identity - ecological consistency analysis: I would suggest to remove the rating of the p-values into "significance"-levels and stick to one alpha value (0.05) throughout the manuscript. What would be interesting is a statistic that provides the degree of association between the two variables tested (like the R statistic in a correlation) - does the fisher test provide that? If not, then it is a test intended as a yes / no test and should only be used this way. You might want to overhaul this whole section in that regard.

When reporting results, please ensure throughout the result section that these are reported as a statement followed by a reference to a table, figure or statistical test. For example, "A was larger than B (Figure 1)", "C was smaller than D (Table 2)" or "E was correlated with F ($R = 0.6$, $p\text{-value} = 0.03$)". Sometimes this was not clear and it dilutes the exciting findings.

The "reflectance curve consistency" section felt perhaps the least tight. I suggest to spent some more time and energy on this. Here, I would have loved to have the differences between the spectra of the habitat types described in more detail as well as the relevant parts of the spectrum highlighted in text (not just in the figure captions). While I follow the reasoning of displaying the data in three different ways, I think this is actually perhaps not that helpful as it dilutes the findings - it makes it difficult to identify what really matters. I'd thus suggest focussing on Figure 3.5 and it's description (what are the key findings), then moving the other figures to the Appendix. You can still refer to them in a sentence or two in the results text and pick them up later in the discussion.

Discussion

Here I felt like there were some sections that needed a bit more work, while others were very well developed and a real joy to read. I especially liked the limitations and conclusion sections. I think your critical thinking and knowledge of the relevant literature really shone through.

As the most important point about the discussion, I would highlight that the objective of a Masters project (and the scientific endeavour in general) is not to find significant relationships (even if we might hope for them). Instead it is about trying to take a step back and being as open as possible when finding out what the answer to an hypothesis is. Sometimes our expectations are met (and that is easy) and sometimes they are not (that can be hard, but is completely okay!). It is therefore important that, when we discuss our findings, we do not portray it in a way that significant results are the "aim" or "goal". This is both for our own sanity (not underselling / talking down what we achieved), but also for showing that we were able to be as open as possible to the outcome of our question. Please revise the discussion section (and check the other sections too), keeping the reporting focussed on what you found (even if you found no relationship). Of course, it is still important that we discuss why a finding did not meet our hypothesis (which your discussion does very nicely!), but we have to be careful not to present it in a way as if we wanted the outcome to be in a certain way. 😊

Some sections in the discussion had a very nice density of citations, putting the findings into the context of the literature, others were a bit sparse. Consider bringing the density up for the latter.

Section titles

The section titles throughout the thesis were descriptive rather than informative, i.e., they stated what was described in the section, not what the highlights / main findings were. In the method section that is perfect, but in the results and discussion the titles could perhaps be used to highlight the key messages of each section.

In text citations

Check that all citations are provided in a consistent format (Harvard style).

Appendices

I did not have time to go through all the appendices (skimmed some). I hope the feedback for the main sections of the manuscript will help revise those as well. I recommend to reach out to a colleague of yours if you'd like to have a second pair of eyes on it.